AHMED ALVEE

Solutions Architect | AI & Automation Engineering
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Summary

Solutions architect who built AWS data pipelines and LLM workflows for clinics, and took them live with customers on site. Led the first overseas deployment at Mesh Bio and a team of 2-3 solutions engineers. AWS Certified Solutions Architect – Associate. Completing an MBA at Rotterdam School of Management.

Skills

Cloud: AWS (Lambda, Step Functions, SQS, SNS, EventBridge), Terraform
AI and LLM: Amazon Bedrock, LangGraph, LangChain, Gemini, Pydantic, ChromaDB, n8n, Claude Code, Cursor
Integration: REST APIs, Postman, HL7, MLLP, DICOM, Microsoft Graph API
Languages and tools: Python, C++, MongoDB, Git, Linux, Qt, ROS

Professional Experience

Mesh Bio Pte Ltd — Singapore, SG
Solutions Architect — 05/2022 – 12/2025
Singapore health-tech company serving private healthcare providers. Its health screening workflow scored patient risk and built screening reports.

- Cut per-submission time from about 1 hour to 6-10 minutes, including human review, for 30+ clinics by building an AI clean-up pipeline from scratch (LangGraph, Claude on Bedrock, Pydantic, ChromaDB) in which a human reviewer checked and submitted every file.
- Cut the time to onboard a new lab and clinic combination from about a week to a day by replacing a hardcoded Lambda per combination with a config-driven AWS pipeline (Lambda, SNS, SQS, EventBridge Pipes, Step Functions, HL7 integration) that handled about 10,000 records an hour and stayed inside Lambda and MongoDB connection limits.
- Kept wrong data out of patient profiles with record count checks and exception handling, and resolved a lab vendor's unannounced system switch in one day.
- Built the MLLP endpoint that receives HL7 from lab vendors and writes files to S3, and handled hourly scheduled files plus on-demand submissions for the lab, radiology and clinic pipelines.
- Turned the first overseas customer into a lighthouse account that opened follow-on regional projects by recommending a headless API-based workflow and leading 3 weeks of on-site deployment.
- Owned the integration APIs that became the distribution partner's primary way to integrate new clients; joined technical sales calls to map customer tech stacks and integration points.
- Hired, trained and supervised a team of 2-3 solutions engineers as they took over and expanded the APIs and pipelines.

Creative Medtech Solutions Pte Ltd — Singapore, SG
Software Engineer — 09/2021 – 04/2022
Singapore medtech company building a robotic system that ablates cancer cells with high intensity focused ultrasound (HIFU).

- Owned the annotation and user-interaction layer of the imaging system for a HIFU cancer-ablation robot: marking and measuring ablation targets, and DICOM export.

Singapore University of Technology and Design — Singapore, SG

Research Assistant — 02/2018 – 08/2021

University research lab building robotics and wearable systems, working with teams of research assistants and PhD students.

- Mapped 2 floors and 5 rooms in under 6 minutes with a 12+ robot system, owning its locomotion code.
- Recognized 7 hand gestures to command unmanned vehicles with an ML pipeline (Python, ROS, StretchSense).
- Contributed C++ and hardware engineering to 2 more robotics builds: a legged mapping robot and a custom PCB for 3D-knitted wearables.

MBA Consulting Project

RSM Living Management Project — Rotterdam, Netherlands

AI Strategy Consultant (Client: Truvalu) — 06/2026 – 07/2026

MBA consulting project for Truvalu, a Netherlands-based impact investor supporting agri-food SMEs in emerging markets.

- Projected about a 45% cut in monitoring time for an impact investor with a 5-stage AI monitoring workflow, designed as part of a 5-person MBA team.
- Engineered and demoed AI tools and agents (Gemini, Claude, Tally.so) spanning all 5 workflow stages, from data clean-up to reporting.
- Co-led the final client presentation to Truvalu leadership.

Education

Rotterdam School of Management, Erasmus University — Rotterdam, Netherlands

Master of Business Administration (MBA) — 01/2026 – 03/2027

Singapore University of Technology and Design — Singapore, SG

BEng (Honours), Engineering Product Development — 05/2014 – 08/2017

Certifications

- AWS Certified Solutions Architect – Associate, Amazon Web Services (10/2023).